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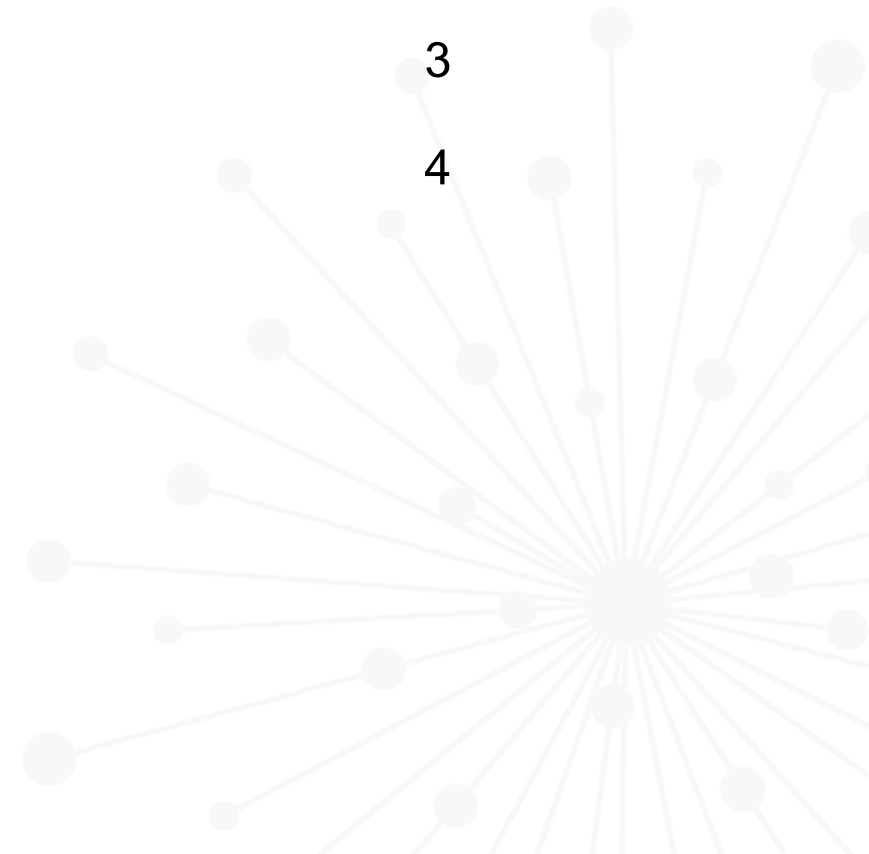
# 3D Gaussian Splatting-based Ground Surface Classification for Humanoid Robot Locomotion Training in Simulation

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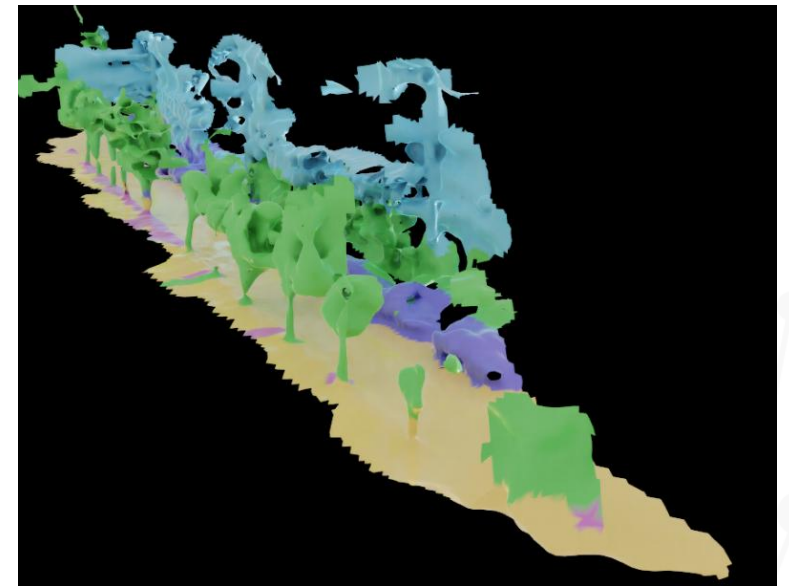
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**The Challenge:** Humanoid robots trained in simulation fail to recognize surface categories and transfer to real environments because real-world terrains have surface-specific physics (friction, restitution, elasticity) that simulators rarely capture.

**The Limitation:** Current 3D reconstruction methods (3DGS, NeRF) recover photorealistic geometry from casual photos but produce unlabeled meshes without semantic information. There is also no automated bridge from visual information (RGB pixels showing grass next to concrete) to physical properties (per-surface friction, restitution, stiffness) which robots can feel.

**Core Problem:** How can we automatically transform a casual single-camera capture of a real outdoor environment into a physics-classified terrain mesh where each ground-surface type drives its own material so humanoid robots can train on faithful, diverse real-world terrain in simulation?



## Limitations of Current Techniques

While articles such as GaussGym implement Gaussian splatting for 3D scene reconstruction and augmented it within the NVIDIA IsaacGym for deep reinforcement learning simulation, they do not cover surface semantic-classification and physics-classified terrain.

### Lack of Semantic Classification

Most 3DGS or real-to-sim pipelines recover photorealistic geometry but produce no per-surface labels, forcing simulators to apply uniform physics across the entire scene.

### Physics real-to-sim 3DGS

Existing semantic 3DGS work stops at rendered video or VR, without producing the per-class collision shapes and material bindings that physics simulators require.

## 3D Scene Reconstruction

**Definition:** Generation of a digital, three-dimensional model of a real-world environment, scene, or object based on data gathered from sensors, such as cameras or LiDAR.

**Role in Research:** Turning multi-view 2D images into a 3D scene model through camera-pose estimation, feature matching, and dense geometry recovery.

## Surface and Mesh Reconstruction

**Definition:** Reconstructing a continuous 3D model from discrete data, such as a 3D point cloud acquired via laser scanners, LIDAR, or photogrammetry.

**Role in Research:** Converting point clouds or Gaussian primitives from 3D scene reconstruction into a triangle mesh.

## 2D Foundation Models for Segmentation

**Definition:** Large-scale artificial intelligence models trained on vast, diverse 2D image datasets (often using self-supervised learning) that can perform or assist in segmenting objects at the pixel level without needing task-specific training..

**Role in Research:** Used together as a three-model supervision for the classifier of scene reconstruction, as well as masking respective objects with classes or labels

## Real-to-Sim RL

**Definition:** Frameworks that bridge real-world scenes into RL simulators where robots train.

**Role in Research:** Utilize the built-in physics engine with classes or labels from segmentation for mesh augmentation to specify each type of materials with its respective physical properties such as stiffness, friction, elasticity etc.

## 3DGS and 3DGS with Segmentation

| Approach                                      | Key Concept                                                                       | Critical Limitation (Research Gap)                                                     |
|-----------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>3DGS</b> (Kerbl et al., ACM ToG'23)        | base method for Gaussian splatting rendering                                      | Geometric outputs and color only, no semantics information segmented by each gaussians |
| <b>Feature 3DGS</b> (Zhou et al., CVPR'24)    | distils 2D SAM features onto each Gaussian, opening 3DGS to feature-based queries | Continuous features with no discrete classes                                           |
| <b>HUGS</b> (Zhou et al., CVPR'24)            | holistic urban semantics with 3DGS                                                | Rendering only with no physical handoffs                                               |
| <b>Gaussian Grouping</b> (Ye et al., ECCV'24) | 16-D identity encoding with joint training                                        | Produces discrete per-Gaussian class IDs                                               |

## Robotics RL and Physics Sim

| Approach                                 | Key Concept                                              | Critical Limitation (Research Gap)                                                                                         |
|------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>VR-GS</b> (Jiang et al., SIGGRAPH'24) | MPM physics directly on Gaussian primitives              | No semantics differentiation which resulted in a single material globally                                                  |
| <b>Pixie</b> (Le et al., 2025)           | supervised learning of 3D physics from pixels            | No per-class collision shape which can't bind physical properties from some Physics engine like PhysX material per surface |
| <b>VISTA</b> (Nagami et al., RA-L'26)    | open-vocabulary semantic GS for online robot exploration | Exploration target only without locomotion training                                                                        |
| <b>Schieber et al.</b> (IEEE VR'25)      | semantics-controlled outdoor GS                          | VR Rendering target without physics simulation                                                                             |

We propose **four components** Reconstruction, Segmentation, Label Transfer, and Sim Integration.

**Reconstruction:** Reconstruct the scene from a casual single-camera capture using existing 3DGS tooling, producing both a Gaussian scene representation and collision mesh

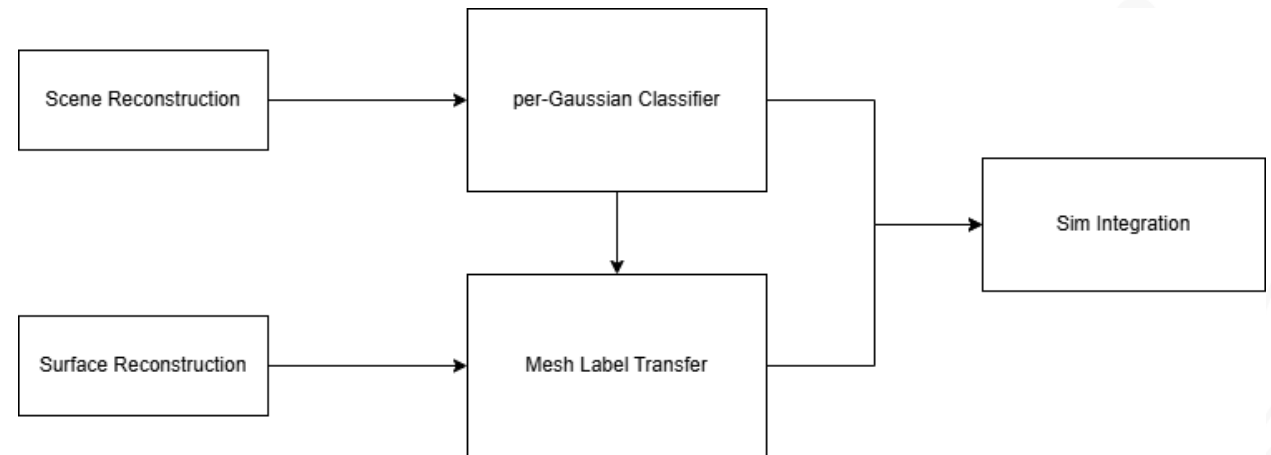
**Segmentation:** For segmentation, we will choose to segment per-Gaussians into groups which will be used to identify materials for each group

**Label Transfer:** Transfer per-Gaussian labels onto the collision mesh's faces

**Bridge to Sim:** Bringing semantic-aware mesh inside for the RL training with existing physics engine and RL tooling, becoming the bridge that turns per-Gaussian labels into a physics-classified mesh (class-aware terrain)

The workflow operates in 4 distinct stages:

- **Reconstruction :**
  - **Scene :** Turning casual single-camera capture into 3D scene
  - **Surface :** Apply existing surface reconstruction technique to gain Mesh from dense point clouds
- **Segmentation :** Utilize existing per-Gaussian classifier method
- **Label Transfer :** Transfer per-Gaussian labels onto mesh faces and apply smoothing
- **Sim Integration :** route per-Gaussian semantic labels into per-class physical materials and train on class aware terrain



## 1.1 Scene Reconstruction

**Input:** Visual observation (RGB-only) in video format

Runs both sparse point clouds (SfM) and fused dense point clouds (MVS)

## 1.2 Surface Reconstruction

**Input:** Fused dense point cloud

Reconstructs a triangle mesh directly from the COLMAP-fused points

## 2.1 2D Mask Supervision

**Input:** RGB frames

Generates per-frame masks and tracks consistent mask IDs across frames

## 2.2 Per-Gaussian Classifier

**Input:** 3DGS scene and temporally-consistent masks

Jointly trains a 16-D identity vector per Gaussian and a classifier head

## 2.3 Class-ID Auto-Labelling

**Input:** Trained classifier and per-class rendered images

Overlap voting maps anonymous class IDs to readable labels

## 3.1 Per-Face Soft Voting

**Input:** Per-Gaussian class probabilities and collision mesh

For each face centroid, the  $k$  nearest Gaussians cast a weighted vote yielding per-face class probabilities

## 3.2 Smoothing

**Input:** Per-face soft labels and face-adjacency graph

Removes single-face outliers, producing the physics-classified mesh.

## 4.1 Mesh Splitting

**Input:** Physics-classified collision mesh

Split per class into different primitives

## 4.2 Physics Material Binding

**Input:** Per-class primitives

Bind rigid body material configuration per primitive

## 4.3 RL Training

**Input:** Class-aware terrain

Train humanoid policy

- **Labelled scene:** collision mesh with per-class configurations and label transfer correctness validated
- **Physics-aware terrain:** demonstration that per-class materials, derived from the bridge, produce a more robust humanoid policy than a uniform-friction baseline

### Metric for Success

We measure per-class face **IoU** and humanoid distance-to-fall improvement over a uniform-friction.

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- **Mesh Label Smoothing**

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